

DRY-RUN GUIDE | 2_pointer.py
Source: Python\5th_Belt\2_pointer.py

CORE BEHAVIOR

Uses two indices to traverse an array, moving one or both based on the comparison shown in the source.

REPRESENTATIVE INPUT

Sorted values [1,2,4,7], target 6 (pair-sum example).

STEP TRACE

(1,7)=8, move right pointer; (1,4)=5, move left; (2,4)=6, match.

EXPECTED OUTPUT / RETURN

Pair (2,4); operation/output depends on this particular source.

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\5th_Belt\2_pointer.py; line numbers refer to the original file. No source was modified.

001 | # Question: Find a pair of numbers whose sum is closest to the target value.

002 | # Add your solution here.

003 |

004 | def solve():

005 | n, target = map(int, input().split())

006 | arr = list(map(int, input().split()))

007 |

008 | i = 0

009 | p = n - 1

010 |

011 | min_diff = float('inf')

012 | best_pair = (0, 0)

013 |

014 | while i < p:

015 | current_sum = arr[i] + arr[p]

016 | current_diff = abs(current_sum - target)

017 |

018 | if current_diff < min_diff:

019 | min_diff = current_diff

020 | best_pair = (arr[i], arr[p])

021 |

022 | elif current_diff == min_diff:

023 | if current_sum < (best_pair[0] + best_pair[1]):

024 | best_pair = (arr[i], arr[p])

025 |

026 | if current_sum < target:

027 | i += 1

028 | elif current_sum > target:

029 | p -= 1

030 | else:

031 | break

032 |

033 | print(best_pair[0], best_pair[1])

034 |

035 | solve()

036 |

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.